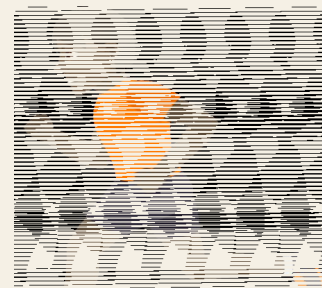
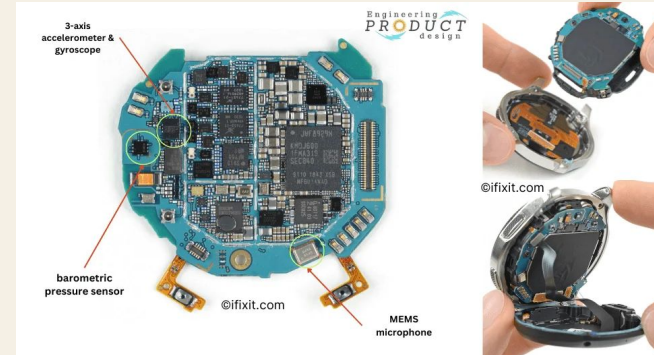
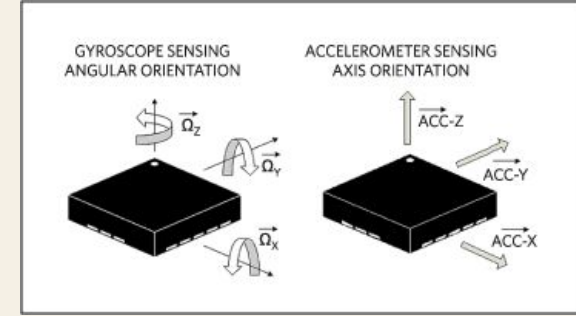


Evaluating Biometric Sensor Accuracy for Human Activity Recognition



Background

- Smartphones and smartwatches have **accelerometers** and **gyroscopes** that continuously collect data. This enables a wide range of applications, including fitness tracking, health monitoring, fall detection,
- Accurate activity classification using these sensors can enhance real-time health insights, improve assistive technologies, and support research in health.



Problem statement:

What is the **most accurate sensor** to use for human activity classification?

Dataset & Description

- <https://archive.ics.uci.edu/dataset/507/wisdm+smartphone+and+smartwatch+activity+and+biometrics+dataset>
- WISDM (**W**ireless **S**ensor **D**ata **M**ining) lab dataset from UCI ML Repository
- Contains raw **accelerometer** and **gyroscope** data from smartphones and smartwatches worn by 51 people performing 18 daily activities
- Activities include walking, jogging, climbing stairs, sitting, and eating
- Contains over **15 million rows**/readings total



EDA

activity	phone_a_count	watch_a_count	phone_g_count	watch_g_count
Total	4804403	3777046	3608635	3440342

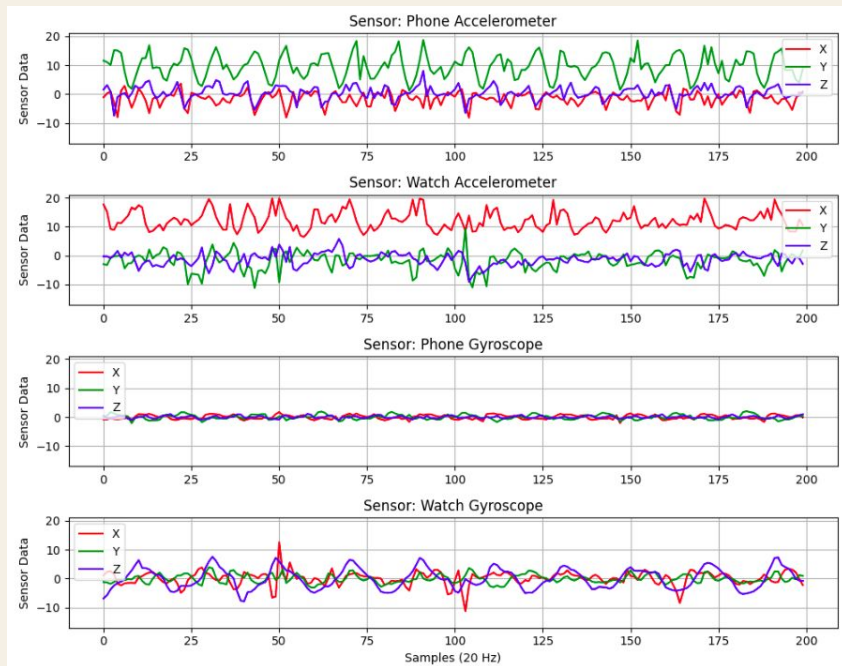
Activity	Code
Walking	A
Jogging	B
Stairs	C
Sitting	D
Standing	E
Typing	F
Brushing Teeth	G
Eating Soup	H
Eating Chips	I
Eating Pasta	J
Drinking from Cup	K
Eating Sandwich	L
Kicking (Soccer Ball)	M
Playing Catch w/Tennis Ball	O
Dribbling (Basketball)	P
Writing	Q
Clapping	R
Folding Clothes	S

activity	phone_a_%	watch_a_%	phone_g_%	watch_g_%
A	5.82	5.57	5.65	5.60
B	5.59	5.45	5.55	5.46
C	5.32	5.49	5.48	5.24
D	5.51	5.64	5.61	5.67
E	5.61	5.73	5.61	5.64
F	5.13	5.43	5.39	5.44
G	5.61	5.53	5.61	5.54
H	5.64	5.55	5.61	5.44
I	5.44	5.56	5.48	5.58
J	5.20	5.38	5.48	5.51
K	5.94	5.72	5.61	5.75
L	5.53	5.39	5.48	5.53
M	5.80	5.55	5.62	5.57
O	5.67	5.56	5.51	5.46
P	5.68	5.63	5.61	5.66
Q	5.42	5.70	5.48	5.74
R	5.58	5.53	5.61	5.55
S	5.52	5.60	5.61	5.62

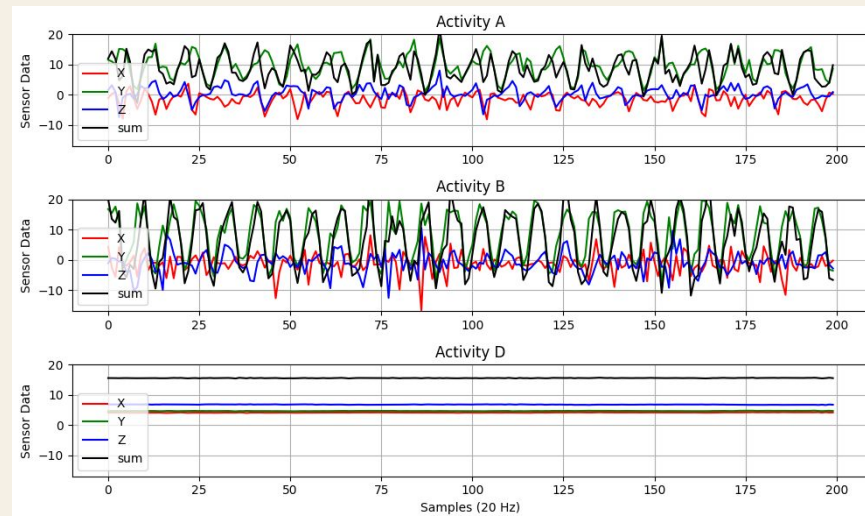
Data cleaning process:

- Renaming columns
- Fixing formatting issues ('z' column had ';' and was a string)
- Sorting data by subject ID, activity type, timestamp
- Segmenting data into 10s windows, each assigned a 'group_id'

EDA (post segmentation)



Different sensors display the patterns for the same activity differently



Signals from different activities have vastly different features

Data Preparation

Feature Engineering:

- Focus on basic statistical features first:
 - Means
 - Standard deviations
 - Correlations
 - Resultant magnitude
- Apply some signal processing
 - Peak finding
 - Discrete (fast) Fourier transform for frequency features

Sensor	# Segments
Phone Accelerometer	23397
Watch Accelerometer	18805
Phone Gyroscope	17396
Watch Gyroscope	17032

Feature Summary

45 features in total

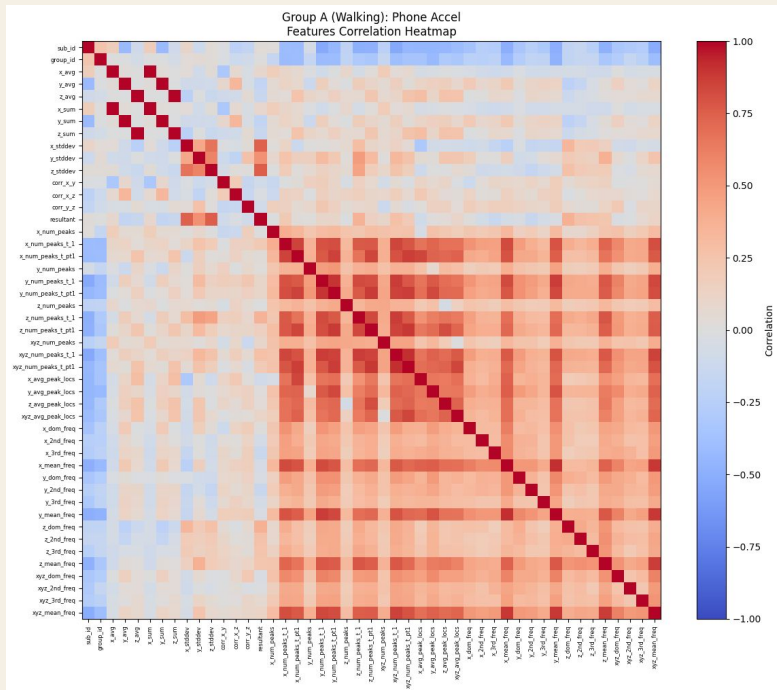
13 statistical (sums, means, corr, resultant)

16 from peak finding

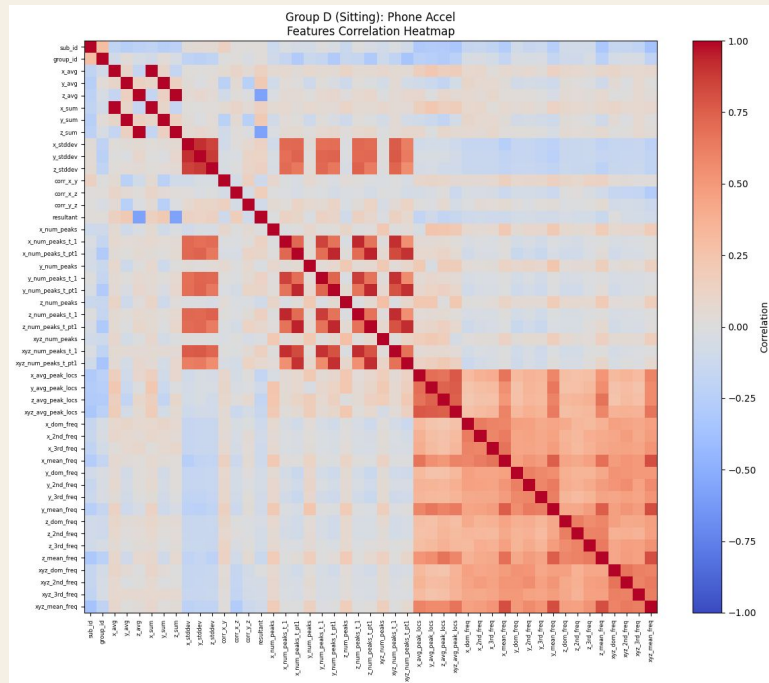
16 frequency related

Feature Correlation Across Activities

Phone Accelerometer: Walking



Phone Accelerometer: Sitting



Analysis Task and Approaches

Task

- What is the **most accurate sensor** to use for human activity classification?
- Phone/Watch Accelerometer and Gyroscope

Approaches

- **Random Forest** classification model
- **Logistic regression** classification model
- Model is trained on data from 50 subjects and tested on 1, this is repeated for each of the 51 test-subjects (using `dask.delayed` and `compute()`)
- Model with all activities, and also with eating activities removed

- ```
rf = RandomForestClassifier(n_estimators=100, max_depth=5, min_samples_leaf=5, min_samples_split=5, random_state=42, n_jobs=-1)
rf.fit(X_train_scaled, y_train)
```

# Analysis Results

## Random Forest (All Activities)

=== Summary of Train and Test Accuracies ===

| Dataset              | Train Accuracy | Test Accuracy |
|----------------------|----------------|---------------|
| features_watch_a.csv | 0.624856       | 0.625979      |
| features_watch_g.csv | 0.591107       | 0.557044      |
| features_phone_a.csv | 0.408191       | 0.314701      |
| features_phone_g.csv | 0.386601       | 0.327061      |

## Logistic Regression

=== Summary of Train and Test Accuracies (Logistic Regression) ===

| Dataset              | Train Accuracy | Test Accuracy |
|----------------------|----------------|---------------|
| features_watch_a.csv | 0.667483       | 0.647068      |
| features_watch_g.csv | 0.642134       | 0.591538      |
| features_phone_a.csv | 0.381429       | 0.307307      |
| features_phone_g.csv | 0.379369       | 0.315748      |

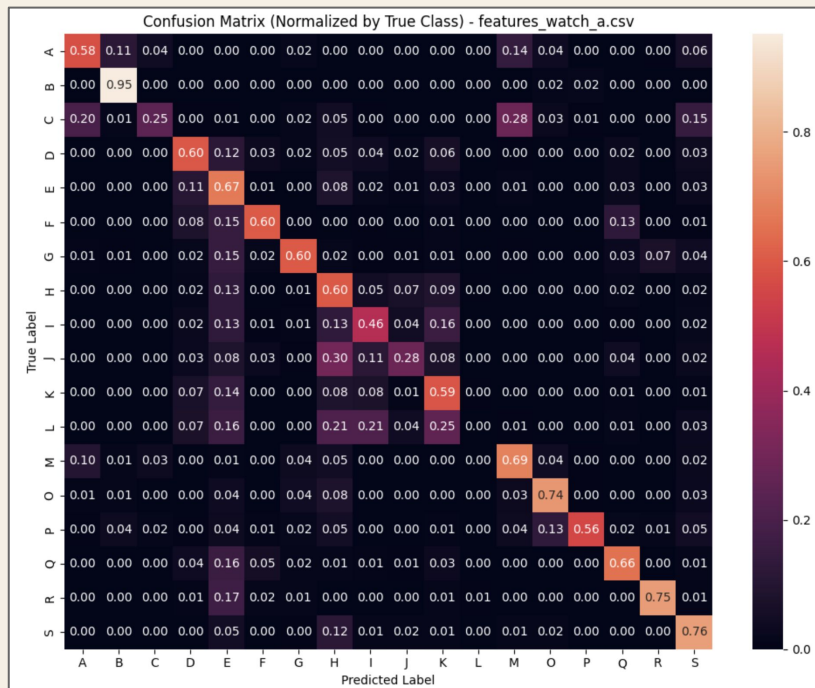
## Random Forest (Minus Eating-Related Activities)

=== Summary of Train and Test Accuracies ===

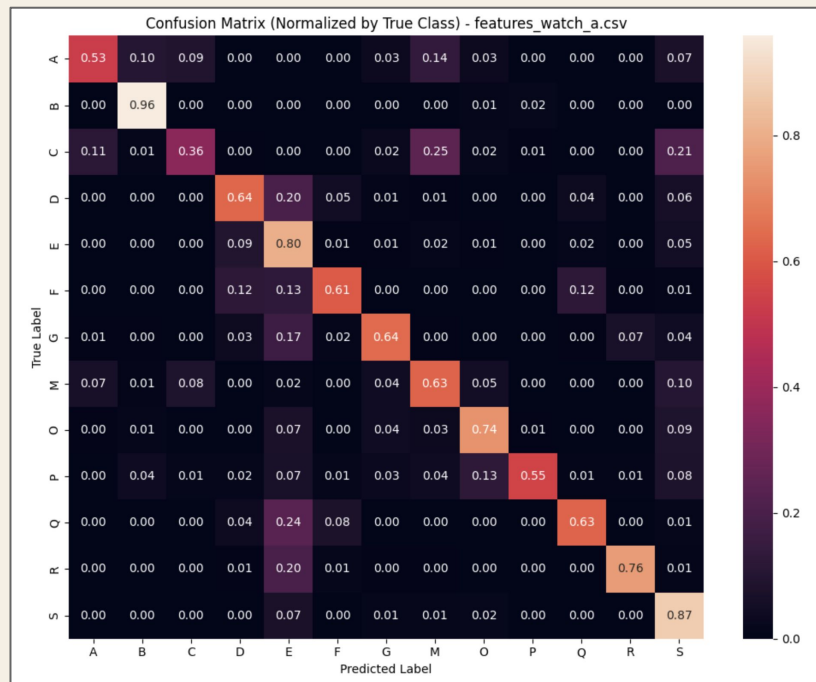
| Dataset              | Train Accuracy | Test Accuracy |
|----------------------|----------------|---------------|
| features_watch_a.csv | 0.739489       | 0.723483      |
| features_watch_g.csv | 0.703221       | 0.664199      |
| features_phone_a.csv | 0.539144       | 0.430581      |
| features_phone_g.csv | 0.515596       | 0.442327      |

# Analysis Results

## Confusion Matrix (RF – All Activities)



## Confusion Matrix (RF – Minus Eating Activities)



# Analysis Results

## Classification Report

### (RF – All Activities)

Average Classification Report (per-class precision, recall, f1-score):

A: Precision: 0.683, Recall: 0.645, F1-score: 0.634

B: Precision: 0.913, Recall: 0.945, F1-score: 0.914

C: Precision: 0.625, Recall: 0.261, F1-score: 0.340

D: Precision: 0.602, Recall: 0.658, F1-score: 0.605

E: Precision: 0.682, Recall: 0.710, F1-score: 0.661

F: Precision: 0.649, Recall: 0.673, F1-score: 0.635

G: Precision: 0.787, Recall: 0.654, F1-score: 0.687

H: Precision: 0.464, Recall: 0.668, F1-score: 0.509

I: Precision: 0.411, Recall: 0.522, F1-score: 0.433

J: Precision: 0.490, Recall: 0.320, F1-score: 0.353

K: Precision: 0.514, Recall: 0.637, F1-score: 0.538

L: Precision: 0.000, Recall: 0.000, F1-score: 0.000

M: Precision: 0.625, Recall: 0.715, F1-score: 0.641

O: Precision: 0.699, Recall: 0.814, F1-score: 0.740

P: Precision: 0.826, Recall: 0.643, F1-score: 0.667

Q: Precision: 0.664, Recall: 0.758, F1-score: 0.681

R: Precision: 0.825, Recall: 0.854, F1-score: 0.823

S: Precision: 0.648, Recall: 0.849, F1-score: 0.702

## Classification Report

### (RF – Minus Eating Activities)

Average Classification Report (per-class precision, recall, f1-score):

A: Precision: 0.756, Recall: 0.586, F1-score: 0.621

B: Precision: 0.916, Recall: 0.953, F1-score: 0.921

C: Precision: 0.608, Recall: 0.377, F1-score: 0.420

D: Precision: 0.768, Recall: 0.703, F1-score: 0.689

E: Precision: 0.756, Recall: 0.821, F1-score: 0.744

F: Precision: 0.692, Recall: 0.679, F1-score: 0.647

G: Precision: 0.807, Recall: 0.709, F1-score: 0.721

M: Precision: 0.616, Recall: 0.652, F1-score: 0.609

O: Precision: 0.714, Recall: 0.813, F1-score: 0.745

P: Precision: 0.802, Recall: 0.633, F1-score: 0.656

Q: Precision: 0.731, Recall: 0.725, F1-score: 0.691

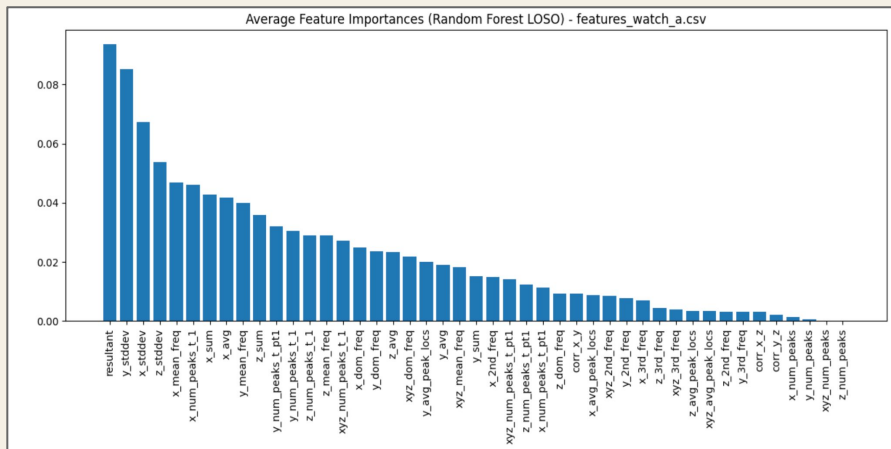
R: Precision: 0.832, Recall: 0.861, F1-score: 0.834

S: Precision: 0.639, Recall: 0.927, F1-score: 0.736

# Analysis Results

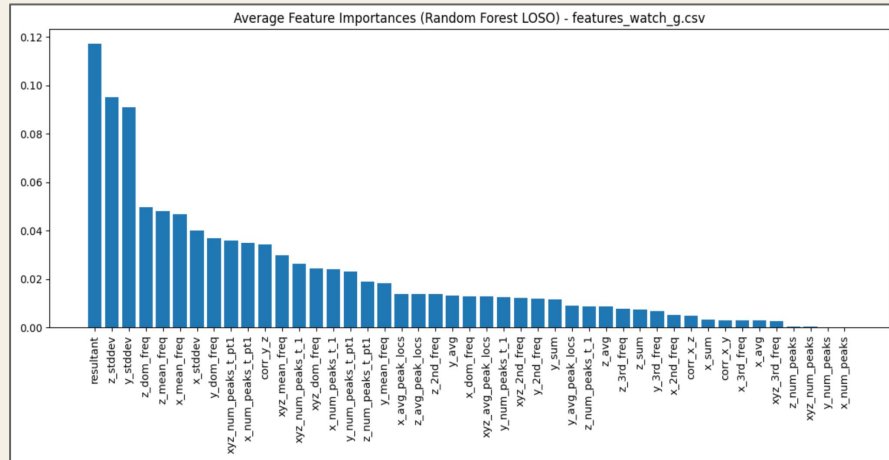
## Feature Importance

(RF – Watch Accel. – All Activities)



## Feature Importance

(RF – Watch Gyro. – All Activities)



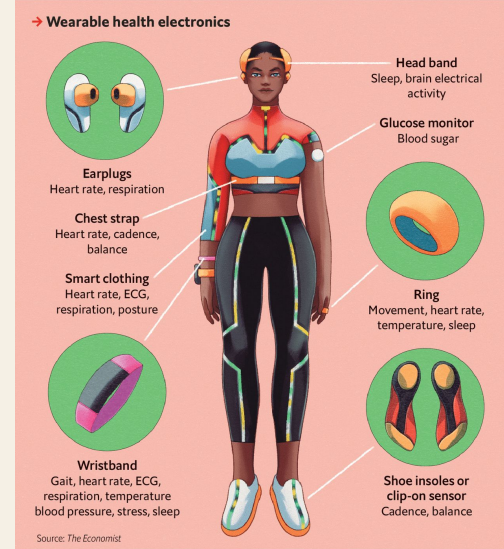
- Resultant and vector stdev are generally the most important, frequency features are more important for gyroscope data

# Challenges and Approaches to Address Challenges

- **Model Overfitting** – parameters for RF model were strict to reduce this, but also negatively impacts accuracy
- **Confusion between eating activities** – compared models with/without these activities
- Model is built on impersonal data, applied to each of the 51 test-subjects, **parallelized using Dask** to reduce computation time/expense
- Raw time-series data – 10s segments were created to allow model training and application

# Insights

- **Watch accelerometer is overall the most accurate sensor for activity recognition compared to the three other sensors**
- Sensor location matters - **phone sensors are inferior to watch sensors** for classification of most activities (in pockets)
- **Eating-related activities are the most difficult to classify** correctly due to potentially less movements overall (eating a sandwich, drinking soup compared to walking, running, etc.)
- Different feature importance for different sensor datasets (mean and other statistics vs. frequency)
- Health sensor industry should focus on **wearable technologies**



# Insights

## 2 versions:

- Spark
- Pandas

```
file = "/watch/gyro/*.txt"
df_watch_g = spark.read.csv(path + file, header=False, inferSchema=True)

print(f"Reading txt files took: {time.perf_counter() - start:.4f} seconds")
```

Reading txt files took: 4.9406 seconds

```
df_watch_g = clean_df(df_watch_g)

print(f"Cleaning dataframes files took: {time.perf_counter() - start:.4f} seconds")
```

rows\_dropped: 0

rows\_dropped: 0

rows\_dropped: 0

[Stage 37:=====] (15 + 2) / 1

rows\_dropped: 0

Cleaning dataframes files took: 9.2427 seconds

```
summary_df = pd.concat([summary_df, pd.DataFrame([tots])], ignore_index=True)

print(f"Summarizing dataframes files took: {time.perf_counter() - start:.4f} seconds")
```

[Stage 93:=====] (16 + 1) / 17

Summarizing dataframes files took: 21.4880 seconds

```
print(f"Generating features took: {time.perf_counter() - start:.4f} seconds")
```

Generating features took: 2.8668 seconds

```
print(f"Sum stats took: {time.perf_counter() - start:.4f} seconds")
```

Sum stats took: 24.7634 seconds

```
df_phone_a = read_all_txt_files("phone/accel/*.txt")
df_phone_g = read_all_txt_files("phone/gyro/*.txt")
df_watch_a = read_all_txt_files("watch/accel/*.txt")
df_watch_g = read_all_txt_files("watch/gyro/*.txt")
```

```
print(f"Reading txt files took: {time.perf_counter() - start:.4f} seconds")
```

Reading txt files took: 6.0193 seconds

```
df_phone_a = clean_df(df_phone_a)
df_phone_g = clean_df(df_phone_g)
df_watch_a = clean_df(df_watch_a)
df_watch_g = clean_df(df_watch_g)
```

```
print(f"Cleaning dataframes files took: {time.perf_counter() - start:.4f} seconds")
```

rows\_dropped: 0

rows\_dropped: 0

rows\_dropped: 0

rows\_dropped: 0

Cleaning dataframes files took: 8.0475 seconds

```
summary_df = pd.concat([summary_df, pd.DataFrame([tots])], ignore_index=True)
```

```
print(f"Summarizing dataframes files took: {time.perf_counter() - start:.4f} seconds")
```

Summarizing dataframes files took: 0.3538 seconds

```
c /= stddev[:, None]
/usr/local/lib/python3.11/dist-packages/numpy/
e
```

```
c /= stddev[None, :]
```

Generating features took: 85.5695 seconds

```
print(f"Sum stats took: {time.perf_counter() - start:.4f} seconds")
```

Sum stats took: 0.0258 seconds

ed UDFs use multiple cores,

x detection run in parallel)

Insights  
upBy().agg().withColumn(...)

efficient with heavy

ion and optimized

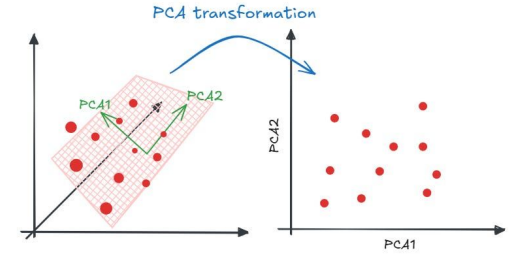
shuffles

| Step                                 | Spark execution time (s) | Pandas execution time (s) |
|--------------------------------------|--------------------------|---------------------------|
| Reading .txt files                   | 4.9406                   | 6.0193                    |
| Cleaning dataframes                  | 9.2427                   | 8.0475                    |
| Summarizing dataframes               | 21.4880                  | 0.3538                    |
| Generating features (UDF/segmenting) | 2.8668                   | 85.5695                   |
| Summarize stats                      | 24.7634                  | 0.0258                    |

# Future Work

- Model overfitting - Dimensionality reduction (PCA), additional features, testing different models and parameters
- Comparing other models, Naive Bayes, K-nearest neighbors, Neural networks (MLP), etc.
- Personal models, trained on data from the target user would be more accurate, and models combining readings from multiple sensors
- Additional Signal Processing and Feature Generation; more filtering, features targeted at individual activities, or using CNN for feature extraction

## Principal Component Analysis



Thank you!